



GE Medical Systems

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Technical Publication

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Revision 00

GE Medical Systems

LS Gantry Pushbutton & Display Flash Kit

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Revision History

Revision	Date	Reason for change
0	11/07/03	Initial Release

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GE M S 以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

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この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本・

非 GEMS 公司的维修员要求非英文本的维修手册时・

客户需自行负责翻译・

未详细阅读和完全了解本手册之前・不得进行维修・

忽略本注意事项会对维修员・操作员或病人造成触

电・机械伤害或其他伤害・

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage that may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Instructions

1 Purpose

This re-flash kit will work:

- To restore code on 2256730, 2256730-2 (Gantry Display), 2256728 (Right Push-button), and 2256729 (Left Push-button) modules that have failed in the field due to lost application code (no longer residing in processor on the circuit board). Any variations or revisions of these part numbers are also included in this (example: 2256728-B) set.
- On modules that are working correctly, to update their boot code to a newer version.

Note: You can tell that a module has lost application code if the green "heartbeat" LED on the module is flashing about 3 times per second. If the module has some kind of unrecoverable error, there is no harm in attempting this procedure, as the module would be discarded anyhow.

2 Effectivity

LightSpeed 2.x - 4.x systems.

3 Required Tools

Qty	Description	Part #
	Laptop with available COM port	
	Flash Cable	2381519
	Flash software: gcflash_package.exe	Download from Support Central http://supportcentral.ge.com/products/sup-products.asp?prod_id=15768

4 Procedure

4.1 Downloading New Flash Code Using gcFlash Tool (Win NT4.0 & 2000)

- 1.) Read these instructions fully before beginning this procedure.
- 2.) Any programs using COM1 should be shut down, unless you have an available COM 2 port and intend to use it. For example "Hot sync", any terminal windows (If you receive "unable to initialize serial port, error 5" you have a COM port conflict)
- 3.) Remove the appropriate gantry covers to gain access to the module(s) you will be flashing
 - Pushbuttons - Side cover(s)
 - Overhead Displays - Side and top covers removed. Front cover still connected but turned such that you can reach the display.
 - Remove power to the gantry (HVDC, AXIAL, 120VAC) using the service switches.
- 4.) If you downloaded the program from Support Central, run the `gcflash_package.exe` file that you just downloaded. The extractor will probably ask if it is ok to create the directory. Click OK.
- 5.) Run the `gcflash` program. If necessary, consult the readme file.

Note: Refer to the "readme" file if necessary. It includes an explanation of when to use the tool.

- 6.) Press ENTER.
Next the preliminary conditions are displayed.
- 7.) Press ENTER.
Now the screen shows a step-by-step procedure for performing the flash download. You can follow the steps on the screen or below:
- 8.) Disconnect the 9-pin cable on the unit to be flashed.
- 9.) Connect the flash cable from your laptop's COM port to the 9-pin connector on the circuit board. Make sure you connect the end labeled "PC Serial" to the PC and the end labeled "Display Control Panel" to the module you intend to re-flash. See [Figure 1](#) or [Figure 2](#) for board connector locations.
- 10.) Turn Dipswitch 1 (on the unit to be flashed) to the ON position. Reference [Figure 1](#) or [Figure 2](#) for dipswitch location.
 - For display units, dipswitch 1 is on the top of the circuit board and is pushed right to turn on.
 - For pushbutton units, dipswitch 1 faces the left side of the circuit board and is pushed up to turn on.
 - Do not touch the other dipswitches. Their settings do not affect the execution of this procedure in any way.
- 11.) Turn on 120 VAC Gantry power using the service switch.
- 12.) Wait for heartbeat LED to blink on module.
- 13.) Press ENTER.

The tool attempts to ping the module. If it is found, you may continue. If there are three failed ping attempts, the program exits.

`gcflash` now sends a command to the module to enter boot mode. This command is 68763k in length. Clicking out of the DOS window may pause the download. If this happens, click in the window to resume.

Next, the board will begin executing the boot code. Both the heartbeat and I/F conn. LED's will be solid green for approximately 10 seconds. During this time, the `gcflash` program attempts to ping the board. When the board is again ready to communicate with the PC, the program allows you to continue.

- 14.) Select 1 for Pushbutton or 2 for Display module.

The tool will now send an application file to the module. This command is 68774k in length. Clicking out of the DOS window may pause the download. If this happens, click in the window to resume.

It should say "Download OK!" If it reads "Download failed" you may try again.

- 15.) Turn off the 120 VAC Gantry power using the service switch.

- 16.) Unplug the PC flash cable from the display / pushbutton unit.

- 17.) Set dipswitch 1 back to OFF.

NOTE: Forgetting to turn dipswitch "Off" leaves module susceptible to noise via spits, other.

- 18.) Reattach the system cable to the display / pushbutton module.

- 19.) Turn the 120 VAC Gantry power back on using the service switch.

When you power the system up, look for "Heartbeat" LED on unit flashing about 2 times per second. This indicates that the board is running in the desired application mode.

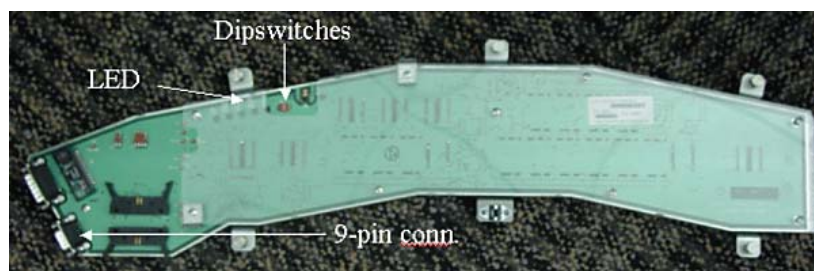
- If it is flashing 3 times per second, this means that the processor is running in boot mode.

- 20.) Once you are done flashing modules on the system, perform a scan hardware reset.

- 21.) Do a functional check of the display / pushbutton module(s).

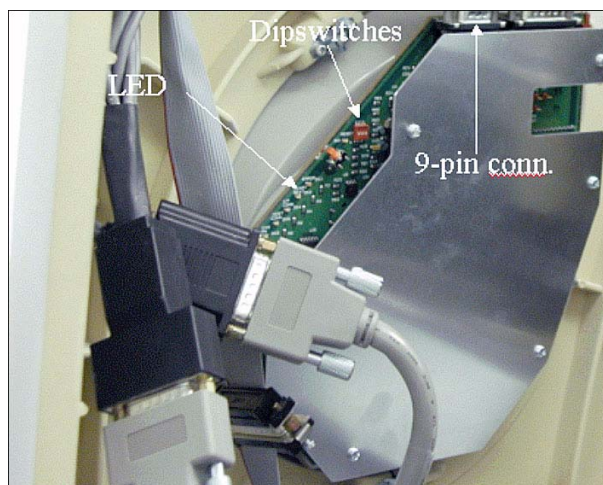
4.2 Reference Illustrations

Figure 1: Gantry Display (Removed) with clear covers.



Note: Many display assemblies will have a metal cover instead of the clear cover shown above. You can still see the LED and read the dipswitch with the metal backing present.

Figure 2: Gantry Right Side Control Panel (on Gantry)



Here you can see the dipswitch and LED locations. On the left modules the layout is similar, just "mirrored." Dipswitches can be either red or black.

4.3 Flash Cable Pinout

PART # 2381519

Should your flash cable labels wear off, the following chart provides the pinout.

Table 1: Flash Cable Pinout

CONTROL PANEL END	PC END	SIGNAL
Pin 1	Pin 5	GND
Pin 3	Pin 3	Tx out of PC
Pin 4	Pin 2	Rx into PC

Note: The pins that are not shown in [Table 1](#) are unused.
Cable style: 9 pin female to 9 pin female, serial style PC cable.
Cable length: 10.5 feet (3.2 meters).

4.4 Known Problems / Mistakes

Table 2: Known Problems / Mistakes

DESCRIPTION	EFFECT	RECOVERY
Flashing pushbutton code to a display or vice versa	Module will not work at functional check	Reflash with correct code using gcflash
Attempting to run program with dipswitch 1 not on	Board is not found	Turn on dipswitch 1 and continue
The code refers to 2256728-2 and 2256729-2 which do not exist.	Do not flash boards with this number if they are found. Currently (10/10/2003), they do not exist.	N/A

End of Instructions



GE MEDICAL SYSTEMS

***GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.***

***GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE***

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